One Real Scalar: Low-Order Induced-Gravity EFT

Jarvis reproducibility artifact

Scope

This report derives and independently validates the local one-loop threshold through curvature-squared order for one real scalar on smooth compact Euclidean backgrounds. It does not claim an expanded B_6 or B_8 operator basis. The source is D.V. Vassilevich, *Heat kernel expansion: user's manual*, electronic PDF p. 40, eqs. (4.26)–(4.28). The companion Markdown report and JSON specification give its local corpus path.

Conventions

$$S_E = \frac{1}{2}\int d^4x\sqrt{g}\,\phi D_E\phi, \qquad D_E = -\nabla^2 + m^2 + \xi R.$$

There is one real scalar, $m > 0$, $d = 4 - 2\epsilon$, and $\overline{\mathrm{MS}}$ subtraction. In the source convention $L = -(\nabla^2 + E)$, so $E = -\xi R$ and $\Omega_{\mu\nu} = 0$.

For mostly-minus Lorentzian signature,

$$g_{\mu\nu} = \eta_{\mu\nu} + \kappa h_{\mu\nu}, \qquad [h_{\mu\nu}] = 1, \qquad [\kappa] = -1.$$

Hence $[\kappa h] = 0$ and $R \sim \kappa\partial^2 h + \cdots$ has dimension two. Therefore B_6/m^2 and B_8/m^4 have Lagrangian dimension four.

Result

With $L_m = \ln(m^2/\mu^2)$,

$$\Gamma^{(1)}_{E,\overline{\mathrm{MS}}} = \int d^4x\sqrt{g}\left[\frac{m^4}{64\pi^2}(L_m - \tfrac{3}{2}) + \frac{m^2}{32\pi^2}(\xi - \tfrac{1}{6})(L_m - 1)R + \frac{L_m}{32\pi^2}B_4 + O(\mathcal{R}^3/m^2)\right],$$

Independent computation and validation

The companion Python script does not merely print the result. First, it encodes the source's general a_2, a_4 tables and uses exact rational arithmetic to substitute $E = -\xi R$ and $\Omega_{\mu\nu} = 0$. Second, it computes the proper-time Gamma integrals at nonzero ϵ, subtracts the $\overline{\mathrm{MS}}$ pole $1/\epsilon - \gamma + \ln(4\pi)$, and Richardson-extrapolates their finite parts. At $m = 2.7, \mu = 1.3$, its absolute finite-part disagreements for B_0, B_2, B_4 are below $9.0 \times 10^{-9}, 2.1 \times 10^{-10}, 3.3 \times 10^{-13}$.

Third, an independent spectral calculation obtains the heat trace on the unit four-sphere from

$$\mathrm{Tr}\, e^{-t(-\Delta + \xi R)} = \sum_{\ell \geq 0} \frac{(2\ell+3)(\ell+2)(\ell+1)}{6} e^{-t[\ell(\ell+3)+12\xi]}.$$

The same code uses the product spectrum $\lambda_\ell = A\ell(\ell{+}1)$, $d_\ell = 2\ell{+}1$ on each factor of $S^2(A){\times}S^2(B)$. A ninth-order forward Lagrange stencil extracts B_2, B_4 from the normalized traces. The three backgrounds S^4, $S^2(1) \times S^2(1)$, and $S^2(1) \times S^2(2)$ then resolve the local R^2, $R^2_{\mu\nu}$, and $R^2_{\mu\nu\rho\sigma}$ coefficients. For $\xi = 0$ and 0.23, those independently recovered coefficients agree with the exact specialization to better than 1.6×10^{-7} and 2.6×10^{-8}, respectively.

The spectral input follows from scalar harmonic homogeneous polynomials on spheres; it is also stated in H. Casini, *Lectures on entanglement in quantum field theory*, exercise 3, p. 42, `https://inspirehep.net/files/19a199a6a26b5847549045339dba86ee`. On every closed validation background the integrated $\Box R$ term vanishes. The script therefore retains that source-specialized coefficient but does not mislabel it as independently spectral-validated.

$$B_2 = (\tfrac{1}{6} - \xi)R,$$

$$B_4 = \frac{1}{2}(\xi - \tfrac{1}{6})^2 R^2 + \frac{1}{180}(R_{\mu\nu\rho\sigma}R^{\mu\nu\rho\sigma} - R_{\mu\nu}R^{\mu\nu}) + \left(\frac{1}{30} - \frac{\xi}{6}\right)\Box R.$$

At $\xi = 1/6$, B_2 and the R^2 coefficient in B_4 vanish. Finite local gravitational counterterms remain adjustable, so these are threshold contributions rather than absolute gravitational couplings.

For a parity-even scalar determinant on a boundaryless background, the local metric expansion has canonical curvature dimensions $0, 2, 4, 6, 8, \ldots$; there is no dimension-nine term. The next prefactors are $-B_6/(32\pi^2 m^2)$ and $-B_8/(32\pi^2 m^4)$. Their expanded tables require a separately declared dimension-six/eight invariant basis.

Reproduction

The companion source is `sakharov_scalar_msbar_low_order.py`. It uses only the Python standard library and emits the source table, exact specialization, raw spectral reconstructions, Gamma-integral finite parts, numerical errors, and all checks.